Unit 3 Mini-Cases

Principles of Capital Markets

Winter Term 2025/26

Please upload your solutions and indicate which Mini-Cases (MCs) you have solved before **November 5, 2025 at 6am**. To do so, go to the assignment section on Canvas and choose "Unit 3 Indication and Submission of MCs". Then answer the true/false questions (Questions 1 to 6) and upload your solutions file (Questions 7). **There will be no deadline extensions! Only complete submissions containing answers to the T/F questions and a solutions upload will be considered, your last complete attempt counts!** You can either write your solutions using software or scan your handwritten solutions. Please note that you may only submit one solutions file. If your solutions consists of more than one file (e.g. 1 pdf and 1 Excel file), please upload a .zip file. Please make sure to hand in only one file of each file type (you might have to merge your scans to one pdf file). Uploaded files will be checked for plagiarism. The solutions in your file should be self-explanatory - not only for the instructor, but for other students as well. Indicate only Mini-Cases you feel confident to present.

Important remark for the presentations: Make sure to provide a concise explanation of your solutions. In particular, when applying any formula from the lecture materials or other sources, you may be asked to explain the structure of your solutions and provide the intuition and underlying concepts behind the formula used.

Mini-Case 3.1 (pdf)

The variance of a return is defined as the expected value of the squared deviation from the mean $var(\tilde{r}) = E[(\tilde{r} - \bar{r})^2]$. However, the variance of a return is also equal to the expected value of the squared return minus the squared mean of the return $var(\tilde{r}) = E[\tilde{r}^2] - \bar{r}^2$.

(a) Show that those two definitions are equal, i.e., $E[(\tilde{r} - \bar{r})^2] = E[\tilde{r}^2] - \bar{r}^2$.

(b) Test this equivalence by calculating both forms of the variance using the returns of security A provided in *Mini-Case 2.1.*

Mini-Case 3.2 (pdf)

Derive a formula for the weights of the minimum variance portfolio of two stocks using the following steps:

(a) Compute the return variance of a portfolio with weights x and $1-x$ on assets 1 and 2, respectively. Show that you get

$$var(\tilde{R}_p) = x^2\sigma_1^2 + (1-x)^2\sigma_2^2 + 2x(1-x)\rho\sigma_1\sigma_2,$$

where $\tilde{R}_p$ is the return of the portfolio, σ_1 is the standard deviation of the return of asset 1, σ_2 is the standard deviation of the return of asset 2, and ρ is the correlation between the returns of assets 1 and 2.

(b) Take the derivative with respect to x of the expression in part *(a)*. Show that the value of x that makes the derivative 0 is

$$x = \frac{\sigma_2^2 - \rho\sigma_1\sigma_2}{\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2}.$$

Mini-Case 3.3 (pdf)

Make use of the following information about the mean returns, standard deviations (both annualized) and correlations for the four asset classes GOLD, STOCKS, REIT and BONDS. Assume that historical data yields unbiased predictors of future performance.

	Correlation with					
Asset	GOLD	STOCKS	REIT	BONDS	Mean return (%)	Standard deviation (%)
GOLD	1	0.10	0.20	0.45	13.14	13.99
STOCKS	0.10	1	0.78	0.12	13.63	15.28
REIT	0.20	0.78	1	0.34	4.26	17.39
BONDS	0.45	0.12	0.34	1	1.24	5.38

(a) Assume you hold a balanced portfolio with a 50/50 allocation across GOLD and REIT, i.e., 50% invested in GOLD and 50% invested in REIT. What is the expected return of your portfolio?

(b) Assume you target an expected return for your portfolio of 9%. What are the portfolio weights you should allocate to GOLD and REIT?

(c) Would an equally weighted portfolio allocated across all asset classes have a higher expected return than the portfolio in *(a)*?

Mini-Case 3.4 (pdf)

Use the information given in *Mini-Case 3.3*.

(a) Assume you hold a portfolio with a 70/30 allocation across GOLD and STOCKS, i.e., 70% invested in GOLD and 30% invested in STOCKS. What is the return standard deviation of your portfolio?

(b) Assume you are considering diversifying into BONDS. What is the correlation of the portfolio in *(a)* with BONDS?

(c) Would an equally weighted portfolio allocated across the three asset classes GOLD, STOCKS, and BONDS have a lower standard deviation than the portfolio in *(a)*?

Mini-Case 3.5 (Excel or R)

Use the information given in *Mini-Case 3.3*. Suppose you are 100% invested in GOLD and want to rebalance your portfolio with weights that will provide a better risk and return combination.

(a) Draw a mean-standard deviation diagram and plot GOLD, STOCKS, REIT and BONDS on this diagram.

You consider investing in one other asset, STOCKS.

(b) Calculate the expected return and standard deviation of multiple two-asset portfolios with different weights for GOLD and STOCKS (e.g., 0% GOLD and 100% STOCKS, 10% GOLD and 90% STOCKS, ..., 100% GOLD and 0% STOCKS)

(c) Plot those portfolios on the mean-standard deviation diagram.

(d) Repeat part *(b)* and *(c)* for the other possible two-asset combinations with GOLD, i.e., GOLD-REIT and GOLD-BONDS.

Mini-Case 3.6 (Excel or R)

Use the information given in *Mini-Case 3.3*. Suppose you are 100% invested in GOLD and want to rebalance your portfolio with weights that will provide a better risk and return combination.

Suppose you would like to earn the highest expected return possible without increasing your initial volatility by allocating some share of your portfolio into STOCKS.

(a) Which portfolio of *Mini-Case 3.5 (b)* consisting of GOLD and STOCKS would you choose?

Instead, suppose you would like to minimize your risk.

(b) Which portfolio consisting of GOLD and STOCKS would you choose? Calculate exact weights.

Hint: The weights of the minimum variance portfolio can be calculated numerically by using, e.g., Excel Solver.
Note, in the case of only two risky assets, the weights of the minimum variance portfolio can also readily be derived analytically (see, e.g., Mini-Case 3.2 (b)).

(c) Repeat part *(b)* for the two-asset combination GOLD-REIT.